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#include <iostream>
#include <iomanip>
#include <limits>
#include <cmath>
using namespace std;

int main() {
   // Representative fare for testing: Rs. 12.34
   constexpr long long FARE_PAISE = 1234;
   constexpr double FARE_RUPEES = 12.34;

   long long journeyCounts[] = {
      20000, 500000, 3000000
   };

   cout << "NAMMA METRO - TOTAL FARE COMPARISON\n";
   cout << "Representative fare: Rs. 12.34\n\n";

   cout << fixed << setprecision(2);

   for (long long journeys : journeyCounts) {
      // Exact integer-paise calculation
      long long totalPaise = journeys * FARE_PAISE;

      // Floating-point calculation
      double totalDouble = journeys * FARE_RUPEES;

      // Compare totals in rupees
```

```
double integerPaiseRupees = totalPaise / 100.0;
double difference =
   totalDouble - integerPaiseRupees;

cout << "Journeys: " << journeys << '\n';

cout << "Integer-paise total: Rs. "
    << integerPaiseRupees << '\n';

cout << "Double total:       Rs. "
    << totalDouble << '\n';

cout << "Difference:         Rs. "
    << difference << '\n';

// Check whether total paise fits in a typical 32-bit int
long long intMax = numeric_limits<int>::max();

if (totalPaise > intMax) {
   cout << "Int paise total: OVERFLOW RISK\n";
} else {
   cout << "Int paise total: "
       << totalPaise << '\n';
}

cout << "Long long paise total: "
    << totalPaise << '\n';
```

```
        cout << "-----------------------------------\n";
    }

    // Demonstrate the limits of int
    cout << "\nDATA TYPE LIMITS\n";
    cout << "Maximum int: "
        << numeric_limits<int>::max() << '\n';

    cout << "Maximum long long: "
        << numeric_limits<long long>::max() << '\n';

    // Demonstrate floating-point representation
    double test = 0.1 + 0.2;

    cout << setprecision(17);
    cout << "\nFloating-point test (0.1 + 0.2): "
        << test << '\n';

    return 0;
}
```